# Tissues I

BIOL2251

Chapter 4

Sections 4.1 – 4.6

# Tissue Type Sorting

# Tissue Type Sorting

- **Instructions**
  - Work on your own (3 min)
    - **Write down everything you remember about each of the four major tissue types**
  - Form small groups (4-6 people)
    - **Share your lists and create one comprehensive group list**
  - Quick show of thumbs (up = confident in all four, sideways = mostly there, down = need more review)
- **Bloom’s taxonomy = Remember**

# Tissue Type Sorting

- **Epithelial:** Covers/lines body surfaces, protection, secretion, absorption, little extracellular matrix, avascular, high regeneration
- **Connective:** Support, connect/anchor, protection, diverse matrix, usually vascularized, cells spread apart
- **Muscle:** Movement, contraction, excitable cells, striated or smooth
- **Nervous:** Communication, impulse transmission, neurons and support cells, excitable

# Epithelial Cell Orientation

# Epithelial Cell Orientation

- **Instructions**
  - Think: **Make a basic diagram of an epithelial cell**
    - **Label: apical surface, basolateral surface, basement membrane. Add at least 2 specializations that might be found on the apical surface**
  - Pair: Compare labels with partner
    - **Why does the apical surface face "free space"?**
    - **What's in the basement membrane and why is it important?**
  - Share: I will call on pairs to share one key function of surface orientation
- **Bloom's taxonomy = Understand, Analyze**

# Epithelial Cell Orientation

- **Apical surface:** Faces lumen/external environment; may have microvilli (absorption), cilia (movement), or glycocalyx (protection)
- **Basolateral surface:** Faces other cells (lateral) and basement membrane (basal); contains cell junctions
- **Basement membrane:** Composed of basal lamina (collagen + glycoproteins from epithelium) and reticular lamina (from connective tissue); anchors epithelium, selective barrier, scaffolding for regeneration
- **Why orientation matters:** Directional transport (absorption/secretion), protection from one side, coordinated function

# Cell Junction Matching

# Cell Junction Matching

- **Instructions**
  - I will give project a list of cell junction types, functions, and real-life analogies
  - **Working in small groups, match the junction type to its function and the analogy**
  - Debrief: Whole class - when might you want each type?
- **Bloom's taxonomy = Remember, Understand**

# Cell Junction Matching

| Junction Type | Function | Analogy |
|---|---|---|
| Tight Junction | Anchor cells together to resist mechanical stress | Zipper / Seal |
| Gap Junction | Allow ions and small molecules to pass directly between cells | Tunnel / Straw |
| Desmosomes | Prevent passage of water and solutes between cells | Rivet / Button snap |

**Tight junctions** → Prevent passage of water and solutes → **Like a zipper/seal**

**Gap junctions** → Allow ions/small molecules to pass directly between cells → **Like a tunnel/straw**

**Desmosomes** → Anchor cells together to resist mechanical stress → **Like a rivet/button snap**

# Epithelial Classification Matrix

# Epithelial Classification Matrix

- **Instructions**
  - Work in groups at your table
  - **Make a grid with cell shapes (squamous, cuboidal, columnar) on one side and layering (simple, stratified) on the other**
    - **For each combination, provide an example function and location**
  - Check your board by comparing with another group
  - Whole class discussion: **Which combinations are most common and why?**
- **Bloom’s taxonomy = Understand, Analyze**

# Epithelial Classification Matrix

| | Simple | Stratified |
|---|---|---|
| Squamous | Diffusion, filtration (lungs, blood vessels) | Protection from abrasion (skin, esophagus) |
| Cuboidal | Secretion, absorption (kidney tubules, glands) | Protection, secretion, absorption (rare – lining some ducts) |
| Columnar | Absorption, secretion (intestines, stomach) | Protection (male urethra, large ducts) |

# Epithelial Tissue Matching Tournament

# Epithelial Tissue Matching Tournament

- **Instructions**
  - Work in small groups – **this is a competition!**
  - **You will receive a list of 6 tissues and 12 scrambled statements**
    - **Match each tissue to 2 statements (1 location, 1 function)**
  - First group to finish holds up paper – all groups stop
  - I will check the answers before you claim your prize
  - We will review correct answers as class
- **Bloom's taxonomy = Remember, Apply**

# Epithelial Tissue Matching Tournament

| Tissues | Locations | Functions |
| --- | --- | --- |
| Simple Squamous | A. Lines most of the digestive tract | G. Stretches to accommodate volume changes |
| Simple Cuboidal | B. Forms air sacs of lungs | H. Diffusion and filtration |
| Simple Columnar | C. Lines the bladder | I. Protection from physical abrasion |
| Stratified Squamous | D. Lines the trachea | J. Absorption and secretion of mucus |
| Transitional | E. Forms outer layer of skin | K. Secretion and absorption |
| Pseudostratified Columnar | F. Forms kidney tubules | L. Moves mucus via cilia |

# Epithelial Tissue Matching Tournament

- **Simple squamous:** B (air sacs/lungs), H (diffusion and filtration)
- **Simple cuboidal:** F (kidney tubules), K (secretion and absorption)
- **Simple columnar:** A (digestive tract), J (absorption and secretion of mucus)
- **Stratified squamous:** E (skin), I (protection from physical abrasion)
- **Transitional:** C (bladder), G (stretches to accommodate volume)
- **Pseudostratified columnar:** D (trachea), L (moves mucus via cilia)

# Gland Type Comparison

# Gland Type Comparison

- **Instructions**
  - Think: **Draw a chart comparing exocrine and endocrine glands**
    - **Include: structure, secretion destination, examples**
  - Pair: Compare with partner, refine your diagram, generate 3 examples of each type
  - Share: Whole class discussion
- **Bloom's taxonomy = Understand, Analyze**

# Gland Type Comparison

**Exocrine Glands:**
- Have ducts that carry secretions
- Secrete onto surfaces or into cavities
- Products: mucus, sweat, oil, digestive enzymes, milk
- Examples: sweat glands, salivary glands, pancreas (exocrine portion), mammary glands

**Endocrine Glands:**
- Ductless
- Secrete hormones into bloodstream/extracellular fluid
- Products travel via blood to distant targets
- Examples: thyroid, pituitary, adrenal, pancreas (endocrine portion - islets)

**Both:**
Derived from epithelial tissue
Secretory function
Can be part of same organ (pancreas)

# Connective Tissue Proper – Matrix Madness

# CT Proper – Matrix Madness

- **Instructions**
  - Work in small groups
  - Answer the following questions:
    - **What are the two main components of connective tissue proper matrix?**
    - **Ground substance: what is it made of and what does it do?**
    - **Fibers: what are the three types and what's special about each?**
    - **How does matrix composition differ between loose and dense CT?**
  - Completion check: Thumbs vote on confidence understanding matrix
- **Bloom's taxonomy = Understand, Analyze**

# CT Proper – Matrix Madness

**Matrix Components:**
- **Ground substance:** Gel-like material between cells and fibers
  - Proteoglycans, glycoproteins, water
  - Functions: holds tissue fluid, medium for nutrient/waste diffusion, resists compression
- **Fibers:** Protein structures providing strength
  - **Collagen:** Strong, rope-like, resists tension (most abundant)
  - **Elastic:** Stretchy, allows recoil after stretching
  - **Reticular:** Fine, branching, forms supportive networks

**Loose vs Dense CT:**
- **Loose:** More ground substance, fewer/looser fibers, more cells, more space
- **Dense:** Less ground substance, tightly packed fibers, fewer cells, less space

# Connective Tissue Proper Mini Cases

# Connective Tissue Proper Mini Cases

- **Instructions**
  - Work in small groups
  - You will be given 4 scenarios
  - For each scenario, **identify the type of connective tissue proper** and explain:
    - **Which tissue is involved?**
    - **What structural features make it suited for this role?**
    - **Where in the body would you find this tissue?**
  - Each group shares on scenario until all 4 scenarios have been review
- **Bloom’s taxonomy = Apply, Analyze, Evaluate**

# Connective Tissue Proper Mini Cases

**Scenario 1:** A patient receives a deep cut that reaches below the skin. The wound is packed with a soft, gel-like tissue that contains many white blood cells fighting infection. This tissue also wraps around blood vessels supplying the area.

- **Answer:** Areolar CT - loose, gel-like matrix, holds tissue fluid, contains wandering immune cells (WBCs), wraps organs/vessels, found under epithelia

**Scenario 2:** During a kidney transplant, surgeons note a delicate mesh-like framework supporting the internal structure of lymph nodes they must navigate around.

- **Answer:** Reticular CT - fine reticular fibers form 3D network, supports lymphoid organs (spleen, lymph nodes, bone marrow), traps cells in framework

# Connective Tissue Proper Mini Cases

**Scenario 3:** A person training for a marathon stores energy reserves that provide insulation and cushion vital organs. This tissue also produces hormones that regulate metabolism.

- **Answer:** Adipose tissue - adipocytes store triglycerides, insulation, protection, energy reserve, endocrine function (leptin), found under skin, around organs

**Scenario 4:** An athlete tears their Achilles tendon, which normally connects calf muscle to heel bone. The tissue is white, tough, and has fibers all running in the same direction.

- **Answer:** Dense regular CT - parallel collagen fibers, withstands tension in one direction, tendons (muscle to bone) and ligaments (bone to bone), white/shiny appearance

# Connective Tissue Proper Mini Cases

**Scenario 5:** A leather craftsman works with animal skin (dermis), noting its toughness and ability to withstand stress from multiple directions. The tissue forms a strong, irregularly arranged layer.

- **Answer:** Dense irregular CT - collagen fibers in multiple directions, withstands tension from various angles, dermis of skin, organ capsules, joint capsules

**Scenario 6:** An infant has difficulty maintaining body temperature because they lack sufficient tissue for insulation underneath their skin.

- **Answer:** Adipose tissue - infants have less subcutaneous fat, develops more with age, brown fat in infants for heat generation

# Tissue Integration Synthesis

# Tissue Integration Synthesis

- **Instructions**
  - Work in small groups
  - Each group will be given an organ, for which you will need to list the following:
    - **Which epithelial tissue(s) line it and why?**
    - **Which connective tissue(s) support it and why?**
    - **What cell junctions would be important and why?**
  - We will share and discuss as a class
- **Bloom's taxonomy = Analyze, Evaluate**

# Tissue Integration Synthesis

**EXAMPLE 1: Small Intestine**

- **Epithelial tissue:**
    - Simple columnar with microvilli and goblet cells
    - Single layer for efficient absorption; microvilli increase surface area

- **Connective tissues:**
    - Areolar CT (lamina propria) - holds blood/lymph vessels for nutrient absorption
    - Dense irregular CT - provides strength for movement/stretching
    - Adipose - cushions and anchors intestines

- **Cell junctions:**
    - Tight junctions - prevent leakage of enzymes/bacteria between cells
    - Desmosomes - hold tissue together during peristalsis
    - Gap junctions (in muscle) - coordinate contractions

# Tissue Integration Synthesis

**EXAMPLE 2: Urinary Bladder**

- **Epithelial tissue:**
  - Transitional epithelium
  - Stretches from thick (empty) to thin (full); impermeable to urine

- **Connective tissues:**
  - Areolar CT - allows stretching, provides elasticity
  - Dense irregular CT - withstands pressure from multiple directions
  - Adipose - cushions bladder

- **Cell junctions:**
  - Tight junctions - prevent toxic urine from leaking into body
  - Desmosomes - hold cells together during extreme stretching
  - Gap junctions (in muscle) - coordinate bladder emptying

# Tissue Integration Synthesis

**EXAMPLE 3: Trachea**

- **Epithelial tissue:**
  - Pseudostratified columnar ciliated with goblet cells
  - Cilia move mucus up and out; goblet cells trap debris/pathogens

- **Connective tissues:**
  - Areolar CT - vascularized to warm/humidify air; allows flexibility
  - Hyaline cartilage (C-shaped rings) - keeps airway open, prevents collapse
  - Dense irregular CT (perichondrium) - covers cartilage

- **Cell junctions:**
  - Tight junctions - prevent pathogens from penetrating tissue
  - Desmosomes - resist mechanical stress from airflow/coughing
  - Gap junctions - coordinate ciliary beating

# Tissue Integration Synthesis

**EXAMPLE 4: Skin**

- **Epithelial tissue:**
  - Stratified squamous (keratinized)
  - Multiple layers for protection; dead surface cells waterproof and resist damage

- **Connective tissues:**
  - Areolar CT (papillary dermis) - capillaries nourish epidermis; sensory receptors
  - Dense irregular CT (reticular dermis) - collagen resists tearing from all directions
  - Adipose (hypodermis) - insulation, energy storage, cushioning

- **Cell junctions:**
  - Desmosomes - extremely abundant; resist mechanical stress and friction
  - Tight junctions - help prevent water loss
  - Gap junctions - coordinate cell communication and responses

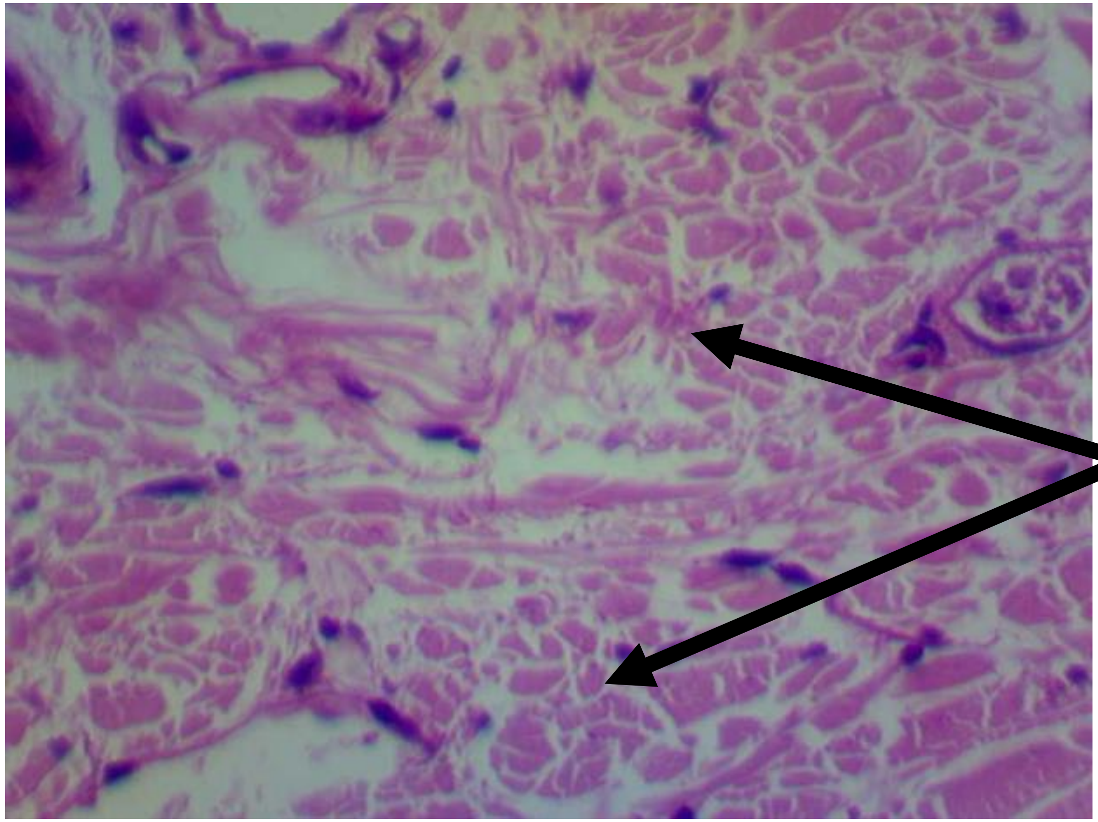
# Tissue Identification

# Instructions

- **Form a team (4-8 people)**
- Identify the tissue presented on the slide
  - 90 seconds to think and discuss as a team
  - Submit your answer on a notecard
    - **Correct tissue identification = 1 point**
    - **Correct function = 1 point**
    - **Correct location = 1 point**
    - **Correct function and location = 3 points**
- **Team with the most points wins!**

# Question 1

# Dense Irregular Tissue (Dense Connective)



No lumen!

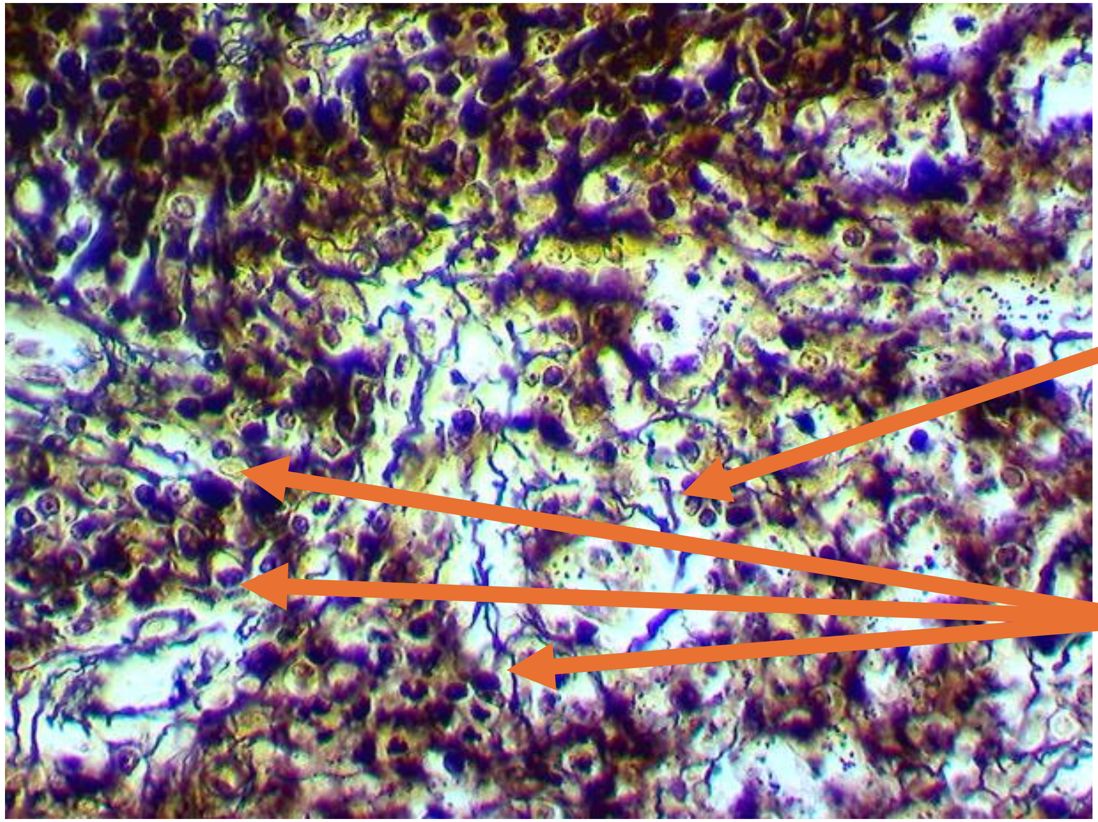
Fibers running in different directions (meshwork)

**Functions**
- Strength to resist forces from many directions
- Helps prevent overexpansion

**Locations**
- Skin
- Periostea
- Perichondria
- Nerve and muscle sheaths
- Organ capsules

# Question 2

## Reticular Tissue (Loose Connective)



No lumen!

Dark, Branching Fibers

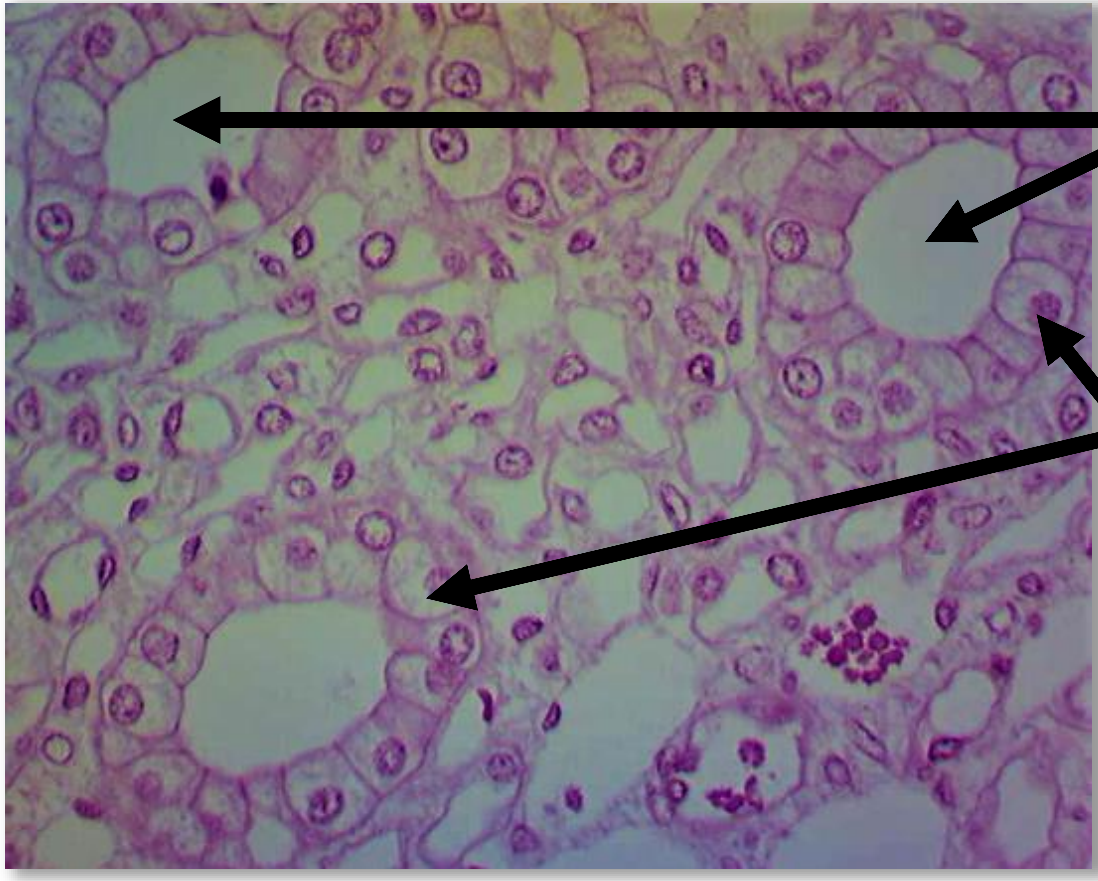
Lots of similar cells

**Functions**
- Supporting framework

**Locations**
- Liver
- Kidney
- Spleen
- Lymph Nodes
- Bone Marrow

# Question 3

# Simple Cuboidal (Epithelial)



Lumen

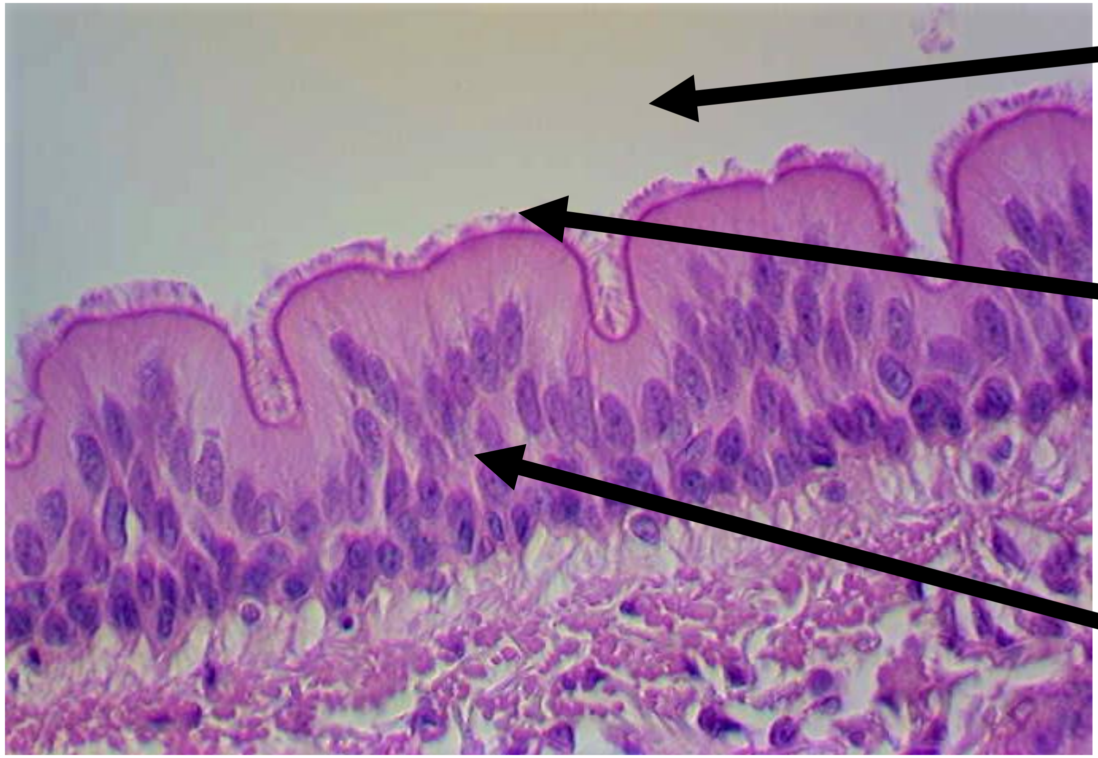
Single layer of square-shaped cells surrounding the lumen

**Functions**
- Secretion
- Absorption

**Locations**
- Glands
- Ducts
- Kidneys
- Thyroid Gland

# Question 4

## Pseudostratified Columnar (Epithelial)



Lumen

Cilia

Tall, Skinny Cells

Nuclei look like they are on several different layers.

**Functions**

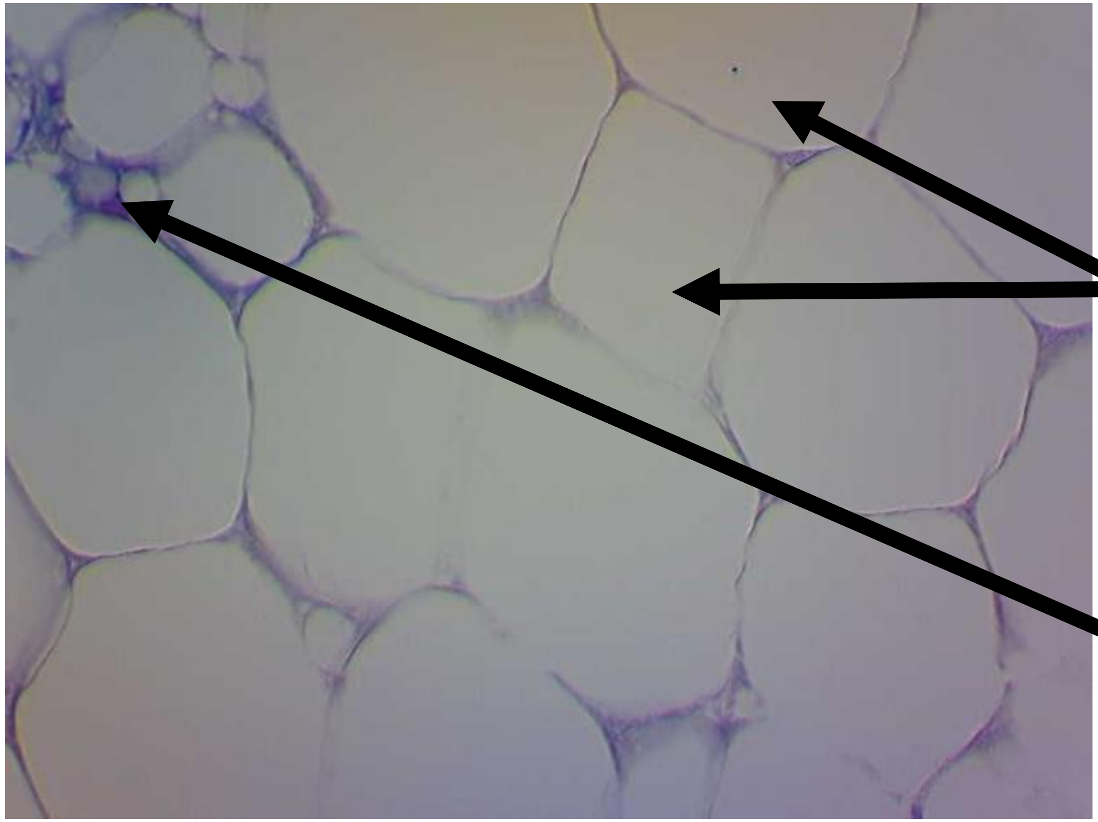
- Secretion
- Protection
- Movement of mucus by cilia

**Locations**

- Lining of nasal cavity
- Trachea
- Bronchi
- Portions of male reproductive tract

# Question 5

## Adipose Tissue (Loose Connective)



No lumen!

Large white blobs

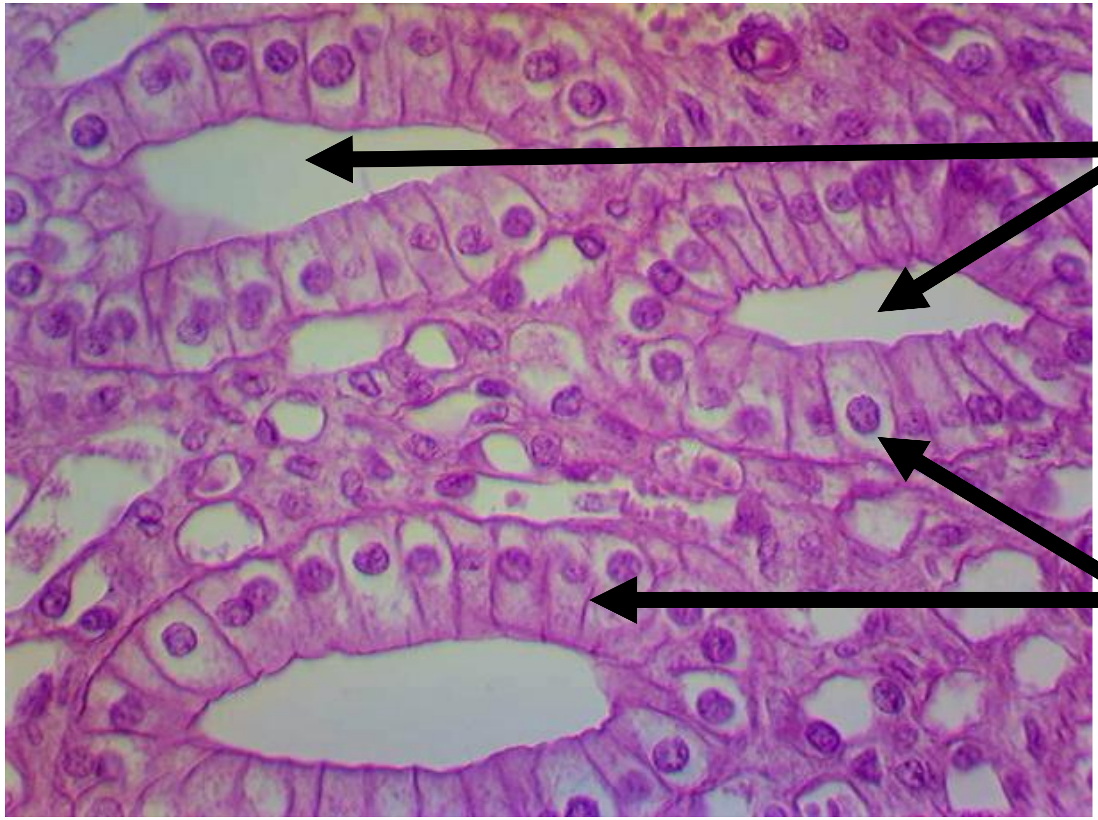
Not many visible nuclei

**Functions**
- Padding
- Cushioning
- Insulation
- Energy Storage

**Locations**
- Under skin
- Buttocks
- Breasts
- Padding around eyes and kidneys

# Question 6

## Simple Columnar (Epithelial)



Lumen

Tall, skinny cells

Nuclei appear to be at roughly the same level

**Functions**

- Protection
- Secretion
- Absorption

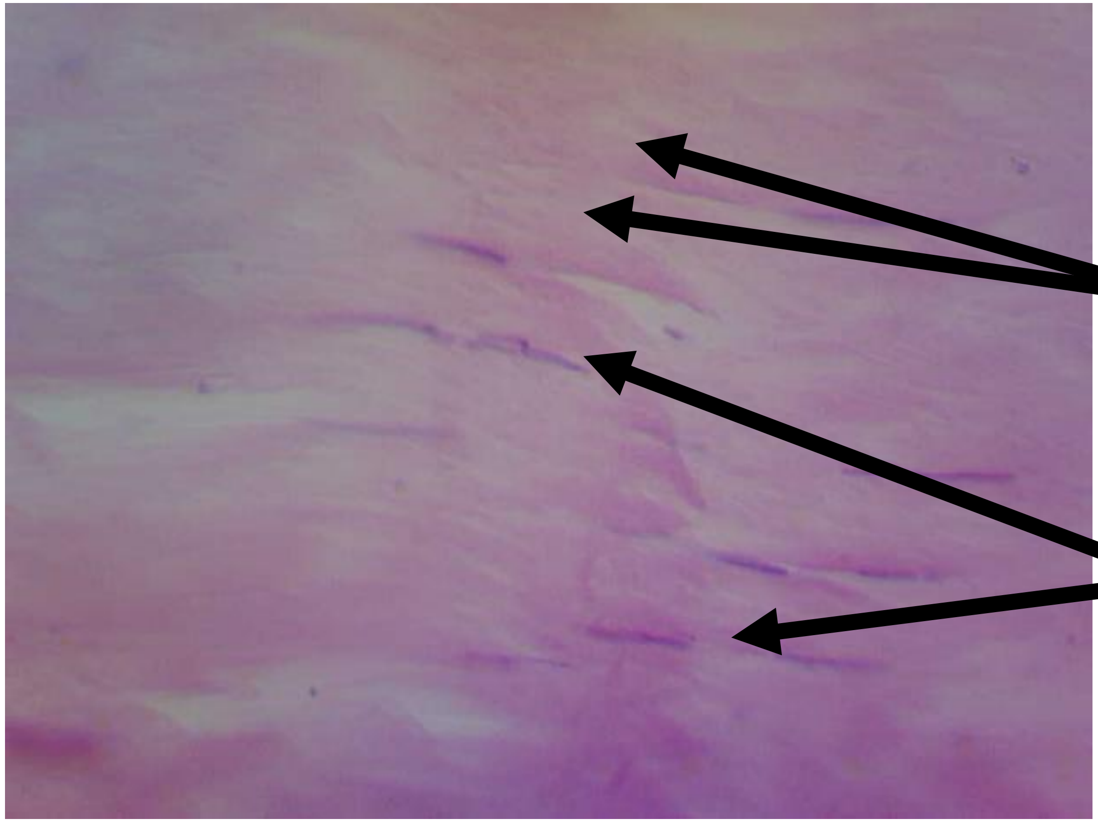
**Locations**

Lining of

- Stomach
- Intestines
- Gallbladder
- Uterine Tubes

# Question 7

## Dense Regular Tissue (Dense Connective)



No lumen!

Parallel Fibers

Cells (fibroblasts) squished between fibers

**Functions**

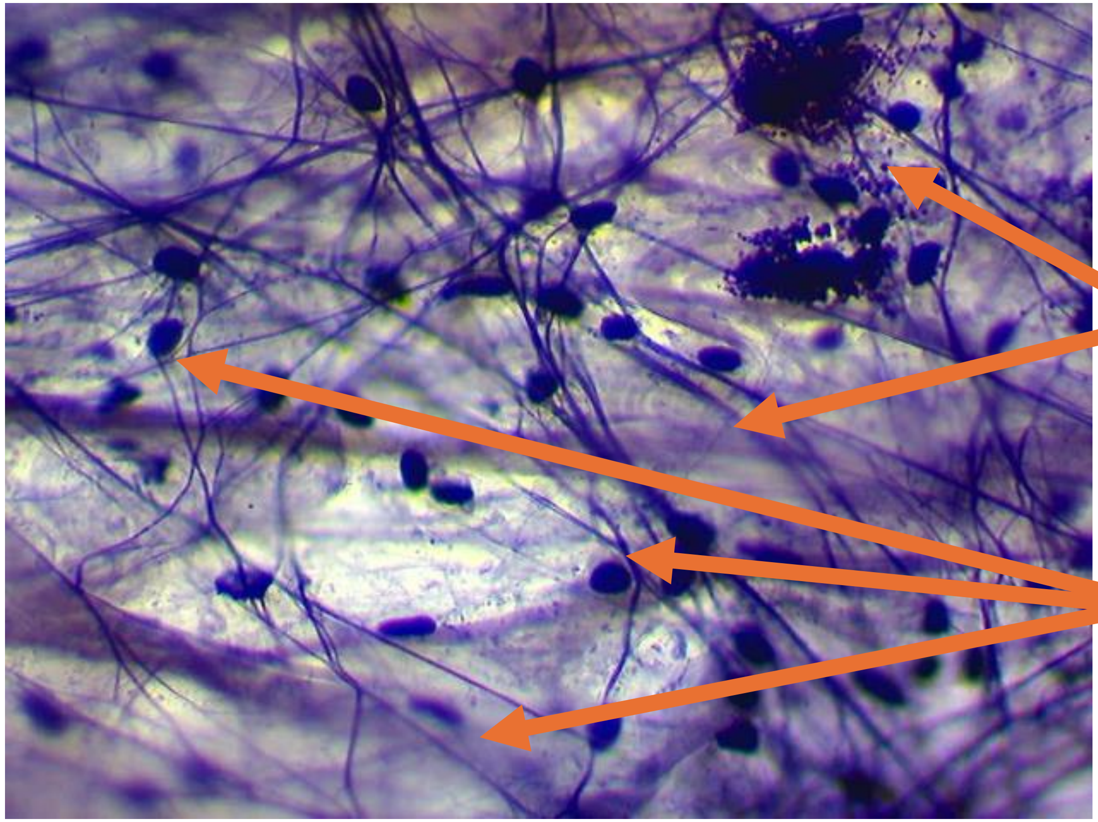
- Firm attachment
- Pull of muscles
- Reduces friction between muscles
- Stabilizes bone positions

**Locations**

- Tendons
- Ligaments
- Covering of skeletal muscles

# Question 8

## Areolar Tissue (Loose Connective)



No lumen!

Scattered cells (many types)

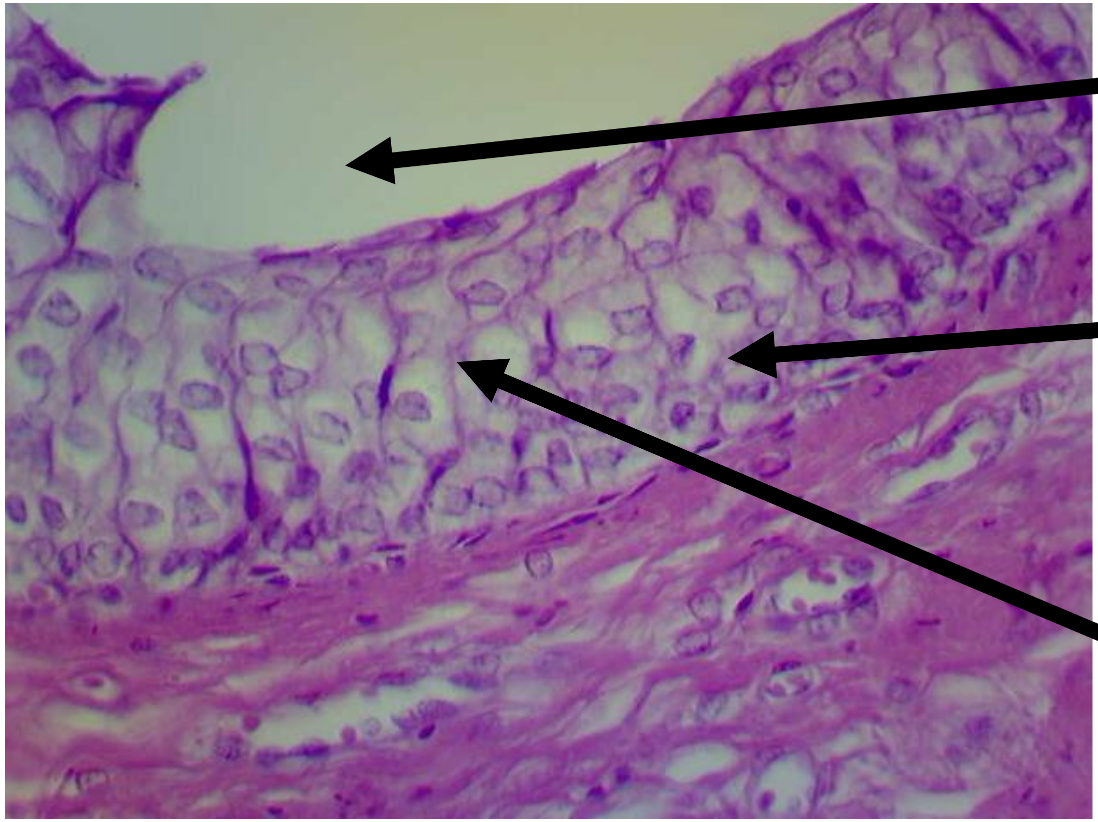
Different fiber types

**Functions**
- Cushions organs
- Support with movement
- Defense (immunity)

**Locations**
- Underneath epithelial tissue

# Question 9

## Transitional Tissue (Epithelial)




**Functions**
- Permits expansion and recoil (stretching)

**Locations**
- Bladder
- Renal pelvis
- Ureters

# Question 10 (Two Tissues)